

Assessing the supply and demand linkage of cultural ecosystem services in a typical county-level city with protected areas in China

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ABSTRACT

The construction of protected areas (PAs) is not only the most effective way to preserve biodiversity resources and the natural heritage, but also provides important cultural ecosystem services (CESs) for satisfying human psychological and spiritual needs and improving human well-being. Knowledge of CESs at the county level is vital for the bottom-up promotion of the overall work of China's PAs. The assessment and mapping of the spatial pattern of CESs, and their supply and demand relationship in PAs and their surrounding areas, are of great significance for high-quality development planning at the county scale. In our study, we developed an integrated approach to mapping the supply and demand of CESs, and then integrated the SolVES model with social media comments to explore the matching relationship between CESs' (aesthetic inspiration, education humanities, and recreation and ecotourism) supply and demand using Wuyishan City, China, with its abundant PAs, as a case study. The results showed that the supply of three types of CESs accounting for approximately 70% of the study area was greater than the demand, and thus these CESs have significant future development prospects. The distribution of the three high-supply CESs and tourist destinations showed roughly the same spatial patterns, in which the high-demand hot spots were identified in the south of Wuyishan City. And the total supply and demand levels inside the PAs were much greater than those outside. Therefore, clarifying the matching relationship between supply and demand of CESs at the county level is not only conducive to promoting the sustainable development of natural resources, but also provides a reference for the planning of county-level protection of PAs under each county's jurisdiction.

1. Introduction

A protected area (PA) is a clearly defined geographical space that is recognized, dedicated, and managed through legal or other effective means to achieve the long-term conservation of natural resources along with the associated ecosystem services and cultural values (Dudley, 2008). Since the establishment of the first nature reserve in 1956, numerous PAs have been established in China, covering forests, grasslands, wetlands, deserts, and marine ecosystems, as well as rare and endangered species and germplasm resources, geological relics, natural landscapes, and important water sources. This is essential to protect biodiversity, maintain ecosystem functions, and ensure national ecological security. The construction of PAs not only is the most effective way to preserve biodiversity resources and the natural heritage, but also provides important cultural ecosystem services (CESs) to meet human

spiritual needs and improve human well-being.

CESs refer to the non-material benefits that humans obtain from ecosystems through spiritual satisfaction, cognitive development, entertainment, and aesthetic experiences (Millennium ecosystem assessment, 2005). The supply and demand of ecosystem services form the dynamic flow of ecosystem services from natural ecosystems to human social systems (Jiang, 2017; Milcu et al., 2022), playing an important role in promoting ecological security and sustainable socio-economic development (Jiang, 2017). CESs can be directly perceived and experienced by humans, and are the outcome of the fusion of various socioecological system elements and the complex relationship between humans and natural ecosystems (Daniel et al., 2012). Assessments of CESs must take into account human demand for these services (Geijzendorffer et al., 2015) while quantifying the ability of ecosystems to provide them (Zhao et al., 2019). However, CESs are difficult to

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measure accurately and have not received sufficient attention from researchers. Therefore, exploring CESs from the perspective of supply and demand can help us to better understand the interaction between ecosystems and human well-being (Wei et al., 2018), and motivate public support for the protection of ecosystems and geographical environments.

As CES is intangible, people's perception of CESs are affected by different cultural needs, spatial-temporal scales, and external conditions (Dai et al., 2019). Therefore, approaches to effectively evaluate the supply and demand of CES is one of the research focuses. For the supply quantification, approaches used include value accounting (Costanza et al., 1997), land use matrix method (Burkhard et al., 2009; Vihervaara et al., 2010), and model simulation (Caro et al., 2020; Meng et al., 2020; Zhao et al., 2019). The models include the integrated valuation of ecosystem services and trade-offs (InVEST) model (Caro et al., 2020), artificial intelligence for ecosystem services (ARIES) model (Martínez-López et al., 2019), and social values for ecosystem services (SolVES) model (Zhao et al., 2019), etc. Compared with the first two models, the SolVES model is advantaged in revealing the relationship between environmental variables and specific observation points. Therefore, the model can provide both the objective evaluation of CES and the quantification of the impact of each variables on CES (Phillips et al., 2006) (Everard et al., 2010; Vilà et al., 2010). In comparison, quantification of demand can be categorized into monetary and non-monetary assessment methods (Cheng et al., 2019), which include the expert scoring (Burkhard et al., 2012a), biophysical index (Cui et al., 2019; He et al., 2019), social media data (Clemente et al., 2019; He et al., 2019; Yoshimura and Hiura, 2017), participatory mapping, and interview-based questionnaires (Peña et al., 2015). For example, a photo questionnaire was used to analyze public preferences for recreational activities in the Basque Country of northern Spain (Peña et al., 2015). The SolVES model was used to quantitatively evaluate the five types of CESs in Cuihua Mountain, and the distribution of supply and demand was analyzed based on the responses to the questionnaire survey and socioeconomic data (Zhao et al., 2019). On the basis of the geotagged photo data, the demand preferences and spatial distribution of four types of CESs were determined, and by analyzing multi-source geospatial data, the supply and demand capabilities were assessed in relation to Shanghai's urban center and suburbs (Bing et al., 2021). The assessment of the matches and mismatches can be achieved by comparing CES supply and demand (Baró et al., 2015; Burkhard et al., 2012b). For example, the supply-demand ratio was proposed to represent the spatial matching of ecosystem services (Li et al., 2016; Lorilla et al., 2019). It has been proved that the supply-demand ratio can be applied for the spatial-temporal comparisons at regional scale. The research scale extends from watersheds (Meng et al., 2020), forests (Bagstad et al., 2016), and PAs (Lorilla et al., 2019) to urban agglomerations (Xiao et al., 2017) and metropolitan areas (He et al., 2019).

Previous studies of the supply and demand of CESs have mainly focused on the evaluation of single indicators of aesthetics, leisure, and entertainment values (He et al., 2019; Xiao et al., 2017), and most studies have only focused on the supply side (Martínez-Harms and Balvanera, 2012; Van Bussel et al., 2020), with a relative lack of comprehensive evaluation of the supply and demand of different CESs. In particular, it is difficult to undertake linkage analysis based on a comprehensive evaluation of the CES supply and demand, PAs and their surrounding areas, and it is impossible to accurately reveal the degree of matching of regional supply and demand.

The county is the basic administrative unit in China. As of 2016, China had a total of 2851 county-level administrative divisions, which accounted for more than 90 % of the total land area and nearly 80 % of the country's total population (Ministry of Civil Affairs, 2017). County is the smallest administrative unit for implementation of national policies in China. Thus, administrative units (e.g., counties, municipalities, or census blocks that the smallest unit of tabulation geography defined by the the United States Census Bureau) better facilitate jurisdictional

resource management or conservation policy analysis and allow for easier interpretation by the general public (Kaminski et al., 2021). It is also the micro-unit and key link in relation to species conservation and ecological civilization construction. Except for National Parks governed by the state, other types of PAs are mostly governed alone or separately by the administrative unit where these PAs are located in, rather than the boundaries of PAs. Therefore, it is important to assess CESs at the county scale to understand PA construction and ecological functions in China.

Wuyishan City is a county-level city under the jurisdiction of Fujian Province in China that has been constructing PAs since 1979. Wuyishan City has numerous unique ecological and humanistic resources, including Mount Wuyi, which is a UNESCO World Cultural and Natural Heritage Site (since 1999) and Wuyishan National Park (WNP), one of the first five national parks officially established in China. It is a typical county-level city with abundant PA types within the county-level jurisdiction. Therefore, clarifying the matching relationship between supply and demand of CESs in Wuyishan City not only is conducive to promoting the sustainable development of natural resources, but also can provide a reference point for the planning of land and space protection in county-level cities with PAs and the high-quality construction of nature reserves with WNP as the main body.

We developed an integrated approach to mapping the CES supply and demand, and selected a typical county-level city with abundant PAs, Wuyishan City, as a case to examine the linkage between the supply and demand of CES. The main purposes of this study are as follows: (1) to assess the CES supply and demand and their matching relationship; and (2) to explore the differences in the relationship between supply and demand of CESs in different types of PAs and their surrounding areas in the county-level administrative unit.

2. Methods

2.1. Study area

Wuyishan City (27°27'–28°04'N, 117°37'–118°19'E) is located northwest of Fujian Province (Fig. 1a), and has a total area of 2798 km². The city includes nine towns or villages, with a total population of approximately 260,000. Mount Wuyi is one of the most outstanding subtropical forests in the world. It is the largest, most representative example of a largely intact forest encompassing the diversity of the Chinese Subtropical Forest and the South Chinese Rainforest, with a high level of plant diversity. It also has outstanding faunal diversity, especially with respect to its reptile, amphibian, and insect species. It contains a series of exceptional archaeological sites, including Han City, established in the 1st century BCE and a number of temples and study centers associated with the birth of NeoConfucianism in the 11th century CE. Here, various types of PAs have been established, including WNP, Wuyishan Wetland Reserve, Wuyishan Scenery District (WSD), Wuyishan Nature Reserve (WNR), Wuyishan Cultural and Natural Heritage (WCNH), State-owned Forest Farm (SFF), and Drinking Water Source Protection Area (Fig. 1b). Most of the PAs are distributed in the mountainous areas in the southwest of Wuyishan City. Their key aim is to provide high-quality CESs and maintain the harmonious coexistence of man and nature. Some of them are partly overlapped as various types of nature PAs are established by different government departments over time. The WNP (maximum area) overlaps with some WFP, WNR, and WCNH.

2.2. Quantifying the supply of CESs

We established a framework for assessing and mapping the supply and demand linkage of CESs in a typical county-level city with abundant PAs (Fig. 2).

Referring to the definition and classification of CES by Millennium ecosystem assessment (2005) and the value system simulated by SolVES

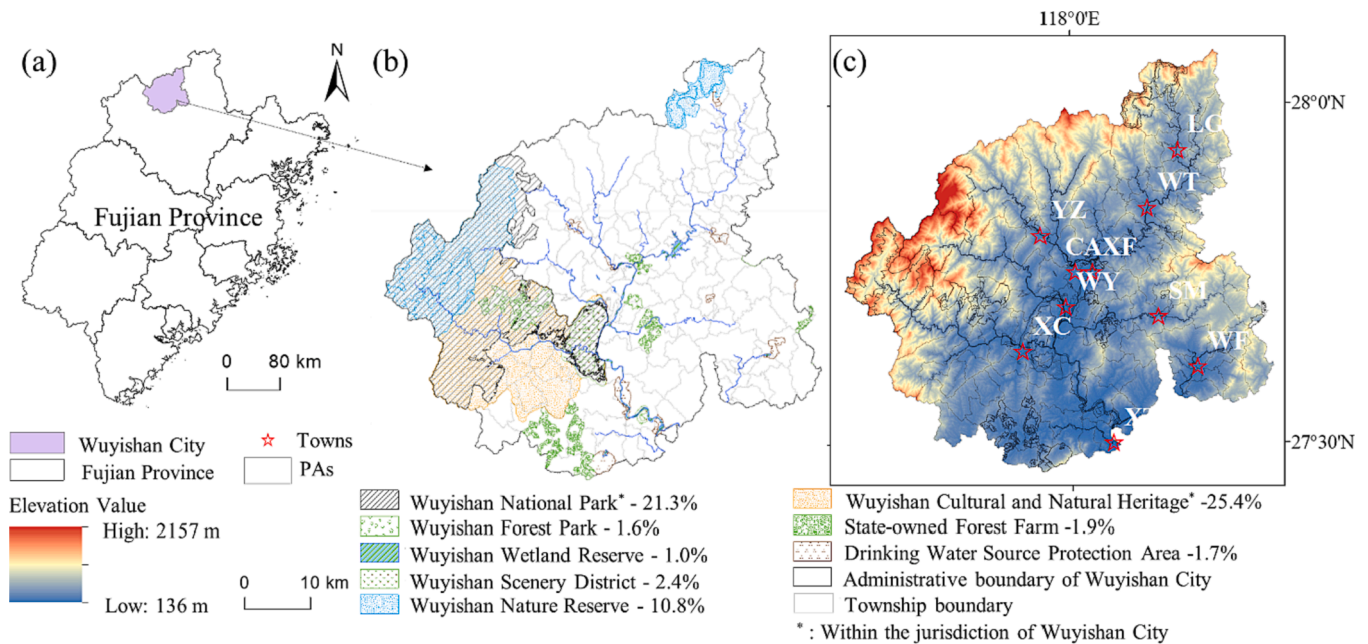


Fig. 1. Geographical location and distribution of PAs in Wuyishan City. The percentages refer to the amount of the total area that is protected. XC: Xincun Town, XT: Xingtian Town, WF: Wufu Town, CA: Chong'an Street, WY: Wuyi Street, XF: Xinfeng Street, SM: Shangmei Village, WT: Wutun Village, LG: Langu Village, YZ: Yangzhuang Village.

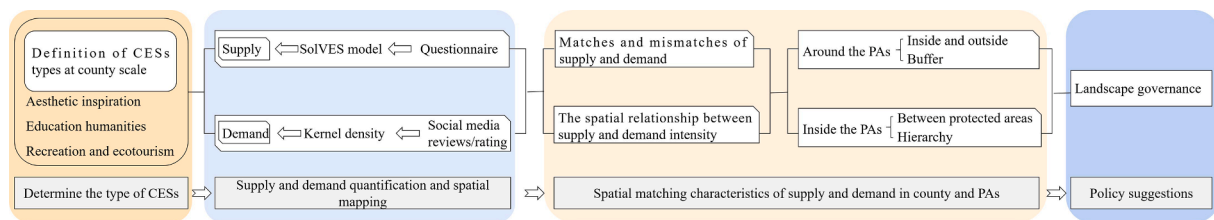


Fig. 2. An integrated approach to mapping supply and demand matching of CESs.

model in previous studies (Sherrouse et al., 2011; 2014), CES can be further divided into more than ten sub-types. In the pre-survey, we adjusted the number of CESs according to the respondents' feedback in the questionnaires. First, respondents suggested considering the characteristics of CESs in the study area. For example, Mount Wuyi is a landscape of great beauty that has been protected for more than twelve centuries, with high aesthetic inspiration value. Mount Wuyi was the cradle of NeoConfucianism, Wuyi palace, Zhizhi Monastery, and numerous cliff carving, and enjoy the education humanities of the Mount Wuyi culture. The spectacular landforms in the eastern scenic area around the Nine-Bend Stream (lower gorge) are of exceptional scenic quality, with isolated, sheer-sided monoliths of the local red sandstone, providing high recreation and ecotourism values. The CESs type should match the natural and human conditions in the study area. Second, there may be more logical causal relationships and overlapping values among subdivided value types. For example, the value of cultural heritage is often synergistic with the educational and historical value. Therefore, various CES are classified into three categories in this study, including aesthetic inspiration, education humanities, and recreation and ecotourism values (Table 1), respectively, to reflect the characteristics of CESs in the study area.

The supply of CESs represents the potential for a specific CES to be offered to people in a specific area at a specific time (Burkhard et al., 2012b). The SolVES model is able to generate a map for visualization of the relationship between CESs values and environment variables (Gao et al., 2017). Thus, we used the SolVES model to quantify the supply capacity of CESs. The SolVES model is a geographic information system

application developed by the United States Geological Survey (USGS) Earth Science and Environmental Change Science Center (Sherrouse and Semmens, 2020). By collecting and analyzing the public's perceived preferences, the value of spatial CESs was able to be quantified and evaluated in a non-monetary manner (using a value index). SolVES model used social survey information as a data source and relevant environmental factors as constraints (Sherrouse et al., 2011). The questionnaires are designed to cover the value classification of various CESs and the value points that could represent CESs supply. Environmental data was used to determine the correlation between high value points of CESs collected through questionnaires and various environmental data. The SolVES model can properly evaluate CES supply as the outcomes of the model are derived from habitat suitability via Maxent (Clemente et al., 2019). In the Maxent approach, environmental data were processed (Phillips et al., 2004; 2006) for evaluating the relationship between social value points and environment, providing CES value index (0–10) map (Sherrouse et al., 2014). In this study, the value index map was used to represent the supply of CES, in which a higher index indicates stronger supply capacity of CES. The maximum entropy model was used to evaluate the model's performance by calculating the area under the curve (AUC) for the ROC, where the training AUC indicated how well the model fitted the study area, and the test AUC represented the value-transfer potential of the model applied to similar areas without survey data. The closer the AUC value is to 1, the better the model simulation effect, enabling the CES supply level to be accurately assessed (Sherrouse et al., 2011).

Questionnaires were distributed in nine towns in the study area to

Table 1

The classification and explanation of ecosystem cultural services in the study area.

Value type	Description	Example	Typical landscapes and main tourist attractions
Aesthetic inspiration	The area provides a rich source of inspiration for activities such as artistic creation, folklore, and architecture, and enables the enjoyment of beauty from different aspects.	Mount Wuyi is the most outstanding area for biodiversity conservation in southeast China and a refuge for a large number of ancient, relict species, many of them endemic to China. The serene beauty of the dramatic gorges of the Nine-Bend Stream, with its numerous temples and monasteries, provided the setting for the development and spread of NeoConfucianism.	
Education humanities	The area has important historical landscapes and objects of important cultural value, which transmit wisdom and knowledge.	Wufu Town retains some ancient buildings, including those from the Song Dynasty. You can experience the thousand-year-old customs and enjoy close contact with the kind local villagers. The trip covers the culture of Zhuxi's teaching academies and provides a deeper understanding of neoConfucian theory and how its flexible teaching integrated with Chinese daily life.	
Recreation and ecotourism	The area is characterized by a natural or man-made landscape, a place where people choose to rest and relax during their leisure time.	After taking the bamboo raft down the stream, you can visit the Nature Museum in the Song Dynasty Historic Street and explore the mystery of the boat-shaped coffins (more than 3,000 years old). You can see an Osmanthus tree more than 900 years old from the Song Dynasty in Wuyi Palace. There are more than 30 stone inscriptions on the cliff of Shuiiguang Rock in a variety of styles, and different meanings.	

(a) The green waters and red mountains of Mount Wuyi. (b) Overlooking the mountains on Great King Peak. (c) Significant numbers of reptile, amphibian and insect species. (d) Wuyi Academy Site. (e) The Zou ancestral hall. (f) Stone inscriptions of Zhu Xi's calligraphy. (g) Xiamei stone carvings. (h) The picturesque Nine-Bend Stream and drifting on the stream. (i) The tea garden looking like fingerprints. (j) The lotus field at Wufu Town. (k) Minyue Kingdom Museum.

enable us to collect the required data. The questionnaire included three components: (1) the demographic characteristics of the respondents, including gender, age, occupation, education, monthly income, and

whether they were residents; (2) respondents' understanding of and interest in cultural service, and the trade-off between ecological environment quality and recreational activities; and (3) we conducted face-

Table 2

Demographic information for survey respondents.

Demographic characteristics	Category	Number of people	Proportion/%	Demographic characteristics	Category	Number of people	Proportion/%
Gender	Male	212	47.64	Interviewee	Tourist	225	50.56
	Female	233	52.36		Resident	220	49.44
Age	≤18 years old	45	10.11	Education	Primary school	20	4.49
	18–24 years old	111	24.94		Junior school	87	19.55
	25–34 years old	128	28.76		Senior school	115	25.84
	35–44 years old	82	18.43		College	202	45.39
	45–54 years old	60	13.48		Master degree/above	21	4.72
	55–64 years old	15	3.37	Income/month (RMB)	less than2,000	151	33.93
	≥65 years old	4	0.90		2,000–5,000	184	41.35
Profession	Businessman	68	15.28		5,000–8,000	69	15.51
	Farmer	40	8.99		8,000–12,000	27	6.07
	Civil servant	56	12.58		12,000–17,000	7	1.57
	Technical staff	54	12.13		17,000–30,000	6	1.35
	Workers	62	13.93		greater than30,000	1	0.22
	Student	115	25.84	Cultural service	Not interested in	11	2.47
	Other	50	11.24		Not really interested in	88	19.78
					A little interested in	227	51.01
Nationality	the Han nationality	406	91.23		Very interested in	119	26.74
	Ethnic minorities	39	8.77				

to-face interviews and distributed questionnaires to investigate respondents' preferences regarding the functional protection of CESs in Wuyishan City. To ensure that the participants' responses were valid, the questionnaires were only completed after the respondents had been given a full explanation of the three CESs by the researchers. In the survey, respondents were first asked to allocate 100 RMB to important cultural service values based on their understanding and preferences (Sherrouse et al., 2011). Then, respondents were asked to identify one to four locations with colored pens on the tourist map providing the service they had chosen. Questionnaires were distributed to households in the study area and to tourists who were encountered in various scenic spots in the WSD, rest areas around the scenic spots, and various townships (Zhao et al., 2022). A total of 492 questionnaires were distributed, and 445 valid responses were received, a response rate of 90.4 %. We also identified 2084 locations for the CES value points. The demographic characteristics of the respondents are presented in Table 2.

The study area is hilly with a forest coverage of 80.5 %, with the elevation from 136 to 2157 m. Besides, since both human disturbances and land use types are closely related to topography, we chose the elevation, slope, and hillshade as the main proxies of environmental variables in our study (Gao et al., 2017; Meng et al., 2020; Zhao et al., 2019). The elevation data using a digital elevation model (DEM) from Geospatial Data Cloud (<https://www.gscloud.cn>), and slope and hillshade data using the DEM to identify the slope and hillshade using spatial analysis tools in ArcGIS. The spatial resolution of all of the raster datasets used was 30 m.

We used the SolVES model to process the questionnaire responses and environmental data and obtain a spatial distribution map of the index values (0–10) of aesthetic inspiration, education humanities, and recreation and ecotourism. To ensure comparability between supply and demand data, we used towns (Micro local government in China) as the smallest unit for CES spatial analysis. Specifically, we summed the index values for each town and divided the total for each town by its land area to obtain the average supply of CESs (Shen et al., 2019). The results were then divided into five supply levels using Jenks natural breaks to represent the supply capacity of different regions.

2.3. Quantifying the demand for CESs

The demand for CESs is an expression of people's preferences for specific CES attributes (Schröter et al., 2014), and can be measured by the number of people seeking CESs and how much they are willing to pay for them (Barkmann et al., 2008). Social media reviews have been widely used in China and elsewhere to assess the aesthetic and entertainment value of locations (Clemente et al., 2019; Yoshimura and Hiura, 2017). As the world's first third-party consumer review website, Dianping.com (the leading mainstream tourist review site in China <https://www.dianping.com>) has a large number of active users who use the platform to freely express their opinions. Users can post reviews, give ratings (0–5), and provide information about the cost to visit a location. After browsing the site, we chose to quantify CES demand by collecting all of the positive reviews and location ratings for the “periphery tour” section (including 15 categories of natural landscapes, cultural relics, theme parks, and other places in Wuyishan City) prior to January 2019.

We undertook a three-step screening process to ensure the quality of the data. First, locations with less than 10 reviews were deleted. Second, comments that were not related to the type of CES and the evaluation of tourist facilities and equipment, such as travel agencies and playgrounds, were deleted. Third, we consolidated recurring comments regarding specific locations. A total of 115 locations were mentioned on Dianping.com, and following the screening process, we identified 56, 49, and 89 locations in relation to aesthetic inspiration, education humanities, and recreation and ecotourism values, respectively. Then, we analyzed the content of the reviews for each location and matched the locations to the three CESs. If the review included perceptions of multiple CESs, the location was listed separately for each type of CES.

Finally, we quantified the level of CES demand based on the number of reviews and ratings for different CESs (Meng et al., 2020).

$$Rp = Lp \times 10 + Cp \times 10 \quad (1)$$

where Rp is the CES demand at location p , Lp is the rating of location p on Dianping.com, and Cp is the number of positive reviews. For consistency in terms of ratings, we assigned values from 1 to 5 based on the number of comments, where 10–100, 101–500, 501–1000, 1001–2000, and more than 2000 comments were assigned values of 1, 2, 3, 4, and 5, respectively. Rp ranged from 0 to 100.

Specifically, we used kernel density to obtain the spatial distribution map of the values for the three CESs and divided the sum of the values of the CES in each town by its land area to obtain the average CES demand. The results were divided into five demand levels using Jenks natural breaks to represent the demand capacity of different regions.

2.4. Assessing matching between the supply and demand of CESs

The ecological supply–demand ratio (ESDR), linking the supply of CESs and human demand, can be used to measure the nature of surpluses or deficits (Li et al., 2016; Lorilla et al., 2019). It was used to reflect the CES supply–demand characteristics in our study area. After normalizing the supply and demand values, we calculated the difference between the supply and demand of CESs in each town as follows (Li et al., 2016):

$$ESDR = \frac{S - D}{(S_{\max} + D_{\max}) \div 2} \quad (2)$$

where $ESDR$ is the CES supply/demand ratio, S and D are the normalized values of the supply and demand of CESs, respectively, and $S_{\max}$ and $D_{\max}$ are the maximum values of the normalized values of the supply and demand of CESs, respectively. $ESDR$ ranges from -1 to 1 . When $ESDR$ is greater than 0, CES supply is greater than demand. When $ESDR$ is approximately equal to 0, CES supply and demand are balanced. When $ESDR$ is less than 0, CES supply is less than demand.

PAs can also conserve natural resources and help to maintain the harmonious coexistence of humans and nature and support sustainable development. To examine the relationship between CESs and human well-being in different types of PAs, we first selected six types of PAs, including WNR, WNP, WSD, WCNH, WFP, and SFF to compare the supply and demand of CESs inside and outside PAs and between PAs. Then, the buffer analysis tool was used to identify the scope of PAs' CESs. The buffer interval is modified by the nearest neighbor distance method, and the 13 km was set to the maximum buffer radius with buffer interval fixed to 2 km (Joppa and Pfaff, 2010), to explore supply and demand from near the PAs to further afield.

3. Results

3.1. The spatial pattern of CES supply

The results of spatial clustering and the maximum index values obtained using the SolVES model are shown in Table 3. It can be seen from Table 3 that the fit of the CESs in the study area was good (AUC of the three types of CESs ≥ 0.93). The results of the average nearest neighbor analysis showed that the spatial distribution of the three CES types belonged to the clustering mode ($R = 0$ and $Z < 0$), and the three CES values were all relatively high (maximum index values ≥ 9), particularly the educational humanities value.

The spatial distribution of the three CES values was approximately identical (Fig. 3). The distribution of high-value areas and tourist destinations showed roughly the same spatial pattern, although the distribution range of aesthetic inspiration values was wider than those of educational humanities values and recreational tourism values. The value index showed a trend of high in the south and low in the north, with multiple scattered cores (Fig. 3a–c). The high-value areas were

Table 3

Index values, spatial clustering, and degree of model fit of the three value types.

The types of CES value	Average nearest neighbor ratio (R)	Standard deviation (Z)	Maximum value index	Mark points	Training AUC	Test AUC
Aesthetic inspiration	0	−49.01	9	657	0.94	0.93
Education humanities	0	−61.09	9	1042	0.94	0.95
Recreation and ecotourism value	0	−48.01	10	630	0.94	0.94

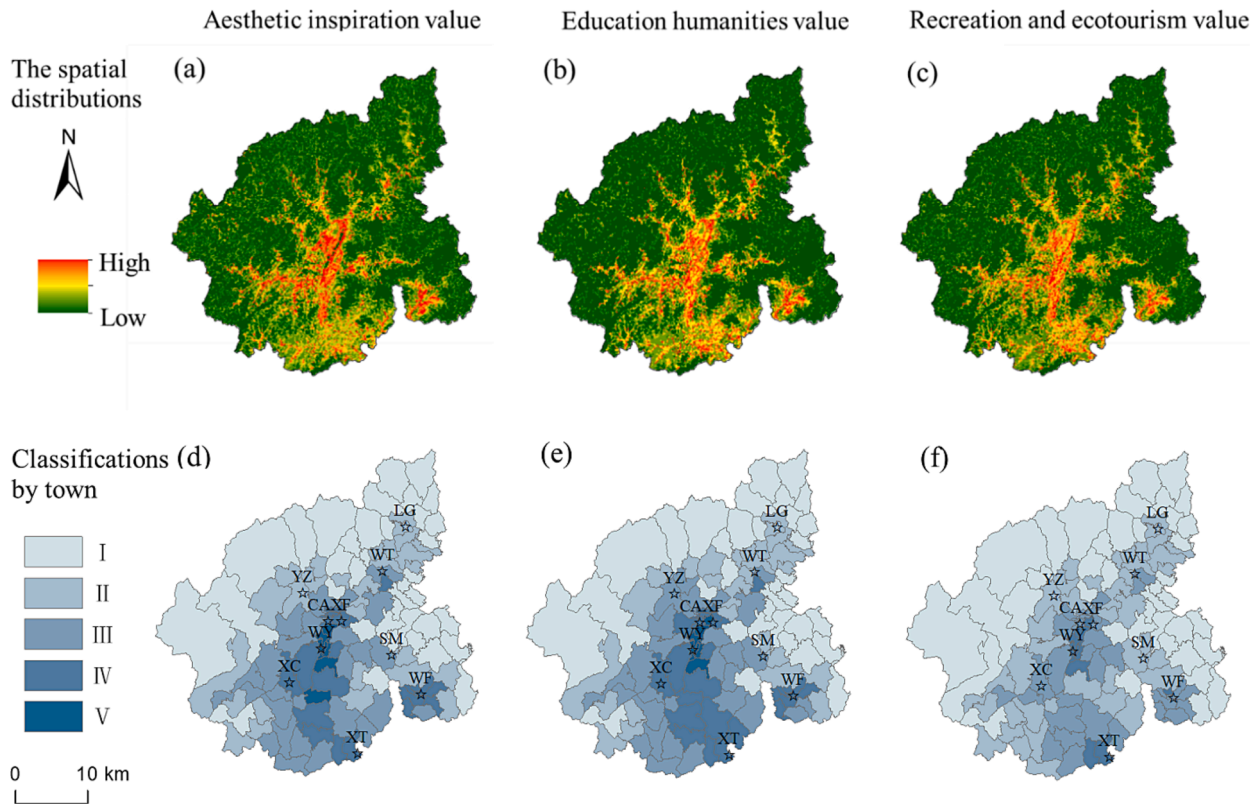


Fig. 3. The supply of cultural ecosystem services: (a), (b), and (c) show the spatial distributions of each service, and (d), (e), and (f) show their classification by town. Class I–Class V: the darker the color, the greater the supply capacity.

concentrated in WSD, WNR, Xingtian Town, and Wufu Town, and there were relatively few high-value points in Shangmei Village, Wutun Village, Langu Village, and Yangzhuang Village in the northern area of Wuyishan City, which has few tourist attractions. A large number of ecological and cultural resources are concentrated in the central area of Wuyishan City. Class IV and V (high-supply) areas account for the smallest proportions, with areas with a high supply of aesthetic inspiration, education humanities, and recreation and ecotourism values accounting for 11.5 %, 14.0 %, and 2.4 %, respectively, of the total area (Fig. 3d–f), while areas with a low supply of recreation and ecotourism values accounted for the largest proportion (80.0 %).

3.2. The spatial pattern of CES demand

The spatial distribution of the three CESs demands were all concentrated in the WSD (Fig. 4a–c). WNP and WCNH are well known, with mature tourism development and excellent infrastructure, and thus are popular tourist destinations. There are few class IV and V (high-demand) areas, and the distribution of high-value areas for education humanities is relatively scattered, while the high-value areas for aesthetic inspiration and recreation and ecotourism are mainly concentrated in WNP. Class I and II (low-demand) areas accounted for a large proportion of the total area, with aesthetic inspiration, education humanities, and recreation and ecotourism accounting for 93.3 %, 94.4 %, and 93.5 %, respectively (Fig. 4d–f). There are few tourist

destinations, such as Wutun Village and Langu Village, in the northern part of Wuyishan City, and the demand for CESs is low.

3.3. Matches and mismatches of supply of and demand for CESs

The proportions of the areas of various levels of supply and demand and the number of corresponding towns in each region showed that approximately 62 % of the regions have sufficient supply capacity (Fig. 5). Fig. 5 shows that the area of high-value supply is larger than that of low-value supply. Only 7.5 % of areas are high-value demand areas. Thus, it is necessary to accelerate the development of tourism resources and the construction of infrastructure to attract people to low-value areas. The supply capacity is greater than the demand capacity, and thus there is significant potential for development in the future.

Fig. 6 shows that the total supply and demand of the three CESs exhibit obvious differences in terms of spatial distribution. Total supply is higher in the south and lower in the north, and there is a wide range of distribution. The greatest amount of supply is concentrated in the WSD and distributed in contiguous areas, while the lowest amount is fragmented and distributed in the Langu Village and Shangmei Village areas (Fig. 6a). Demand is high in the center and low in the surrounding areas and the high-value areas are concentrated in the WSD (Fig. 6b). Total supply and demand figures show that the supply capacity of CESs in most regions (approximately 70.7 %) exceeds demand (Fig. 6c). Only a few areas in Xinfeng Street and Chong'an Street are extremely

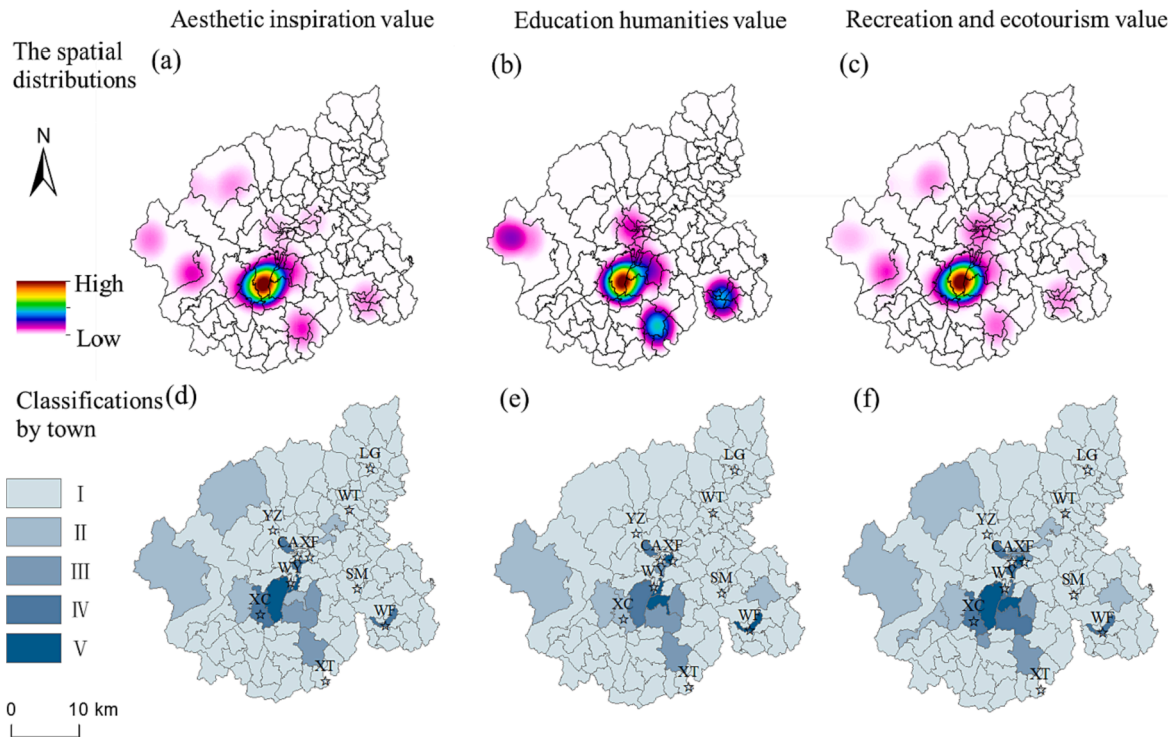


Fig. 4. Demand for cultural ecosystem services: (a), (b), and (c) show the spatial distribution of each service, while (d), (e), and (f) show their classifications by town. Class I–Class V: the darker the color, the greater the demand capacity.

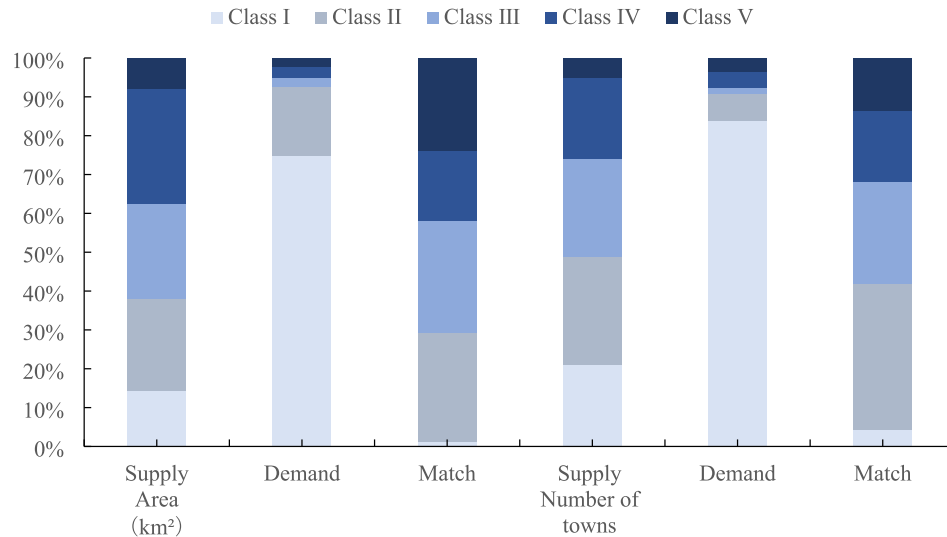


Fig. 5. Proportions of areas with different levels of supply and demand and proportions of the corresponding numbers of towns. Supply and demand: the higher the demand or supply intensity, the darker the color, and the lower the demand or supply intensity, the lighter the color. Matching: the darker the color, the greater the amount by which supply exceeds demand, and the lighter the color, the less the amount by which supply exceeds demand.

unbalanced in terms of supply and demand.

3.4. Supply of and demand for CESs in PAs

The CES supply and demand relationship between the areas inside and outside the PAs (Fig. 7a) shows that the total supply and demand values inside PAs are much higher than those outside PAs. The total supply level of WCNH is the highest among the six types of PAs. The natural resources are highly representative, with rich biodiversity and many historical and cultural relics, and can provide multiple types of CESs. Supply of CESs exceeds demand in all PAs. The level of aggregate

supply is in the order of WCNH > SFF > WSD > WNP > WFP > WNR, while the level of aggregate demand is in the order of WSD > WCNH > WNP > SFF > WFP > WNR (Fig. 7b). The supply and demand capacity of national PAs is greater than that of provincial PAs (Fig. 7c). SFF has the largest supply per unit area, and WSD has the largest demand per unit area (Fig. 7d).

The buffer radius (Fig. 8a) indicates that the continuous buffer areas mainly extend from PA to non-PA areas. Both supply and demand generally decreased with increases in the buffer distance, but the declining trend in demand is significantly greater than that in supply (Fig. 8b). Thus, the gap between supply and demand gradually increases

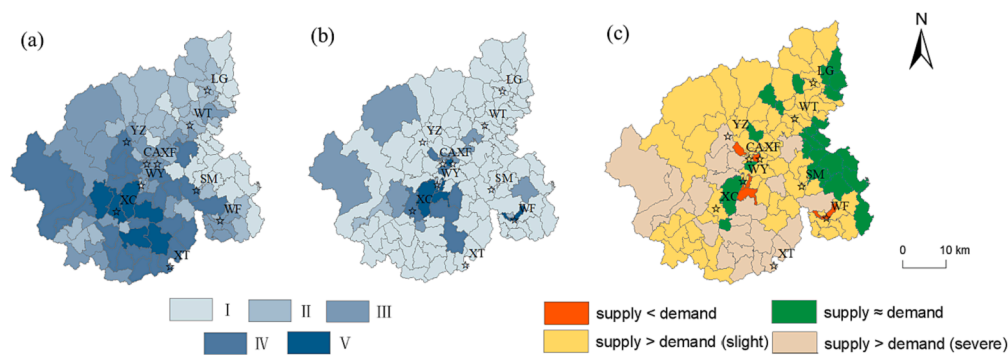


Fig. 6. Spatial distribution of supply and demand matching. (a) Supply. Class I: very low supply area, Class II: low supply area, Class III: moderate supply area, Class IV: high supply area, Class V: very high supply area. (b) Demand. Class I: very low demand area, Class II: low demand area, Class III: moderate demand area, Class IV: high demand area, Class V: very high demand area. (c) Supply/demand ratio ($ESDR$). Jenks natural breaks were used to divide the spatial distribution of supply and demand matching into four levels. Orange indicates $ESDR$ ranging from -0.751 to -0.035 , CES supply is less than demand. Green indicates $ESDR$ ranging from -0.035 to 0.035 , CES supply and demand are balanced. Khaki indicates $ESDR$ ranging from 0.035 to 0.302 , and gray indicates $ESDR$ ranging from 0.302 to 0.760 , CES supply is greater than demand. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

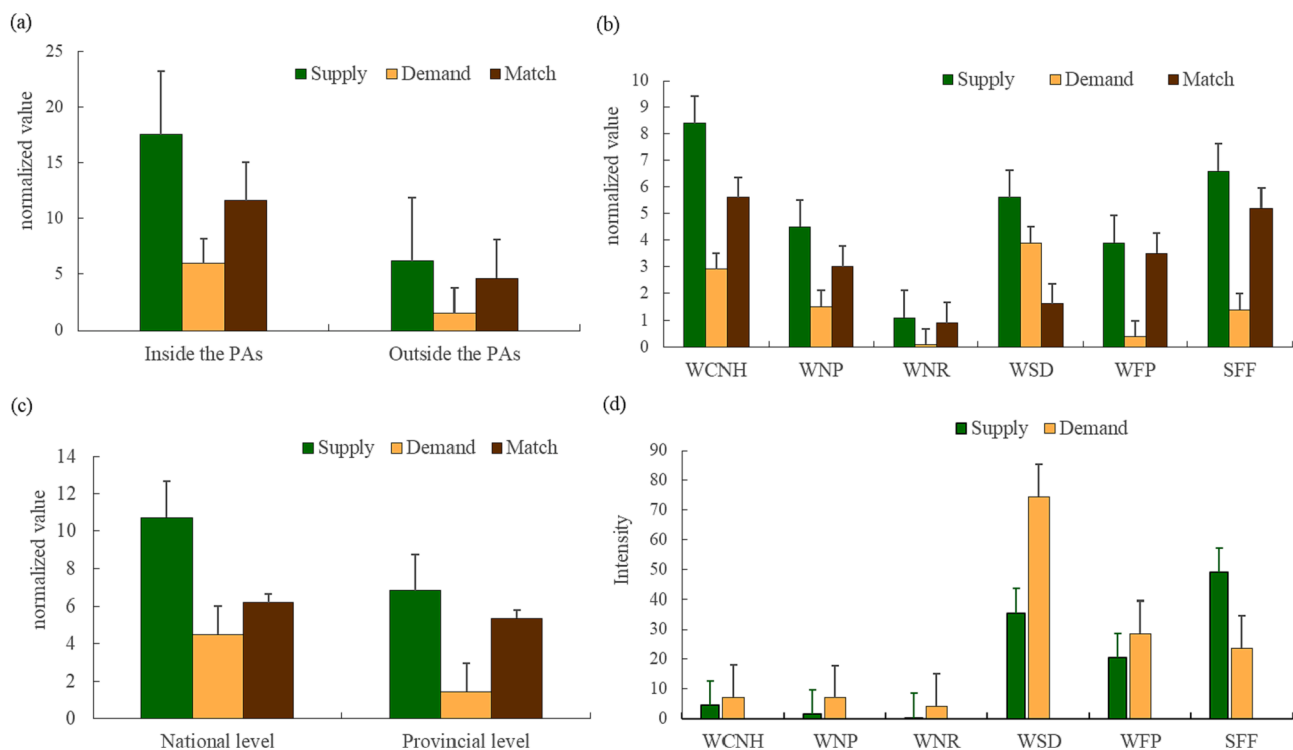


Fig. 7. Relationship between the inside and outside of the PAs in Wuyishan City and the relationship between the PAs. (a) The relationship between the inside and outside of the PAs. (b) Normalized values of supply and demand for each PA. (c) Supply and demand relationships at the national and provincial levels. (d) The ratio of the supply and demand value of each PA to the area of each PA. Intensity = Supply and demand value of each PA/area of each PA. Drinking water source PAs and wetland PAs were too small and scattered to be considered.

within the 7 km buffer zone and then starts to narrow until the 11 km buffer zone.

4. Discussion

4.1. The spatial distribution of CES supply and demand

Understanding the distribution of supply and demand enables us to better determine the spatial availability of various CESs in the region (Zhao et al., 2019). Our study used a spatially explicit approach by integrating the SolVES model with social media comments to explore the matching relationship between CESs' supply and demand. Compared with the traditional method, the method used here can cover all areas studied with high efficiency. The results indicated that the spatial

distribution of supply was similar for the three CES values, and a large number of ecological and cultural resources were concentrated in the central area of Wuyishan City. The distribution of high-value areas and tourist destinations followed roughly the same spatial patterns. This is because tourist areas can provide visitors with beautiful natural scenery and diverse ecological resources, increasing their perceived value of CESs. The results were consistent with previous studies (Kalinauskas et al., 2022).

The supply of CESs is related to specific landscape attributes (e.g. aesthetic attributes, biophysical attributes, and humanistic attributes, etc.) and regional infrastructure construction status. The southern region has more tourism resources, it contains a series of exceptional archaeological sites, including the Han City established in the 1st century BCE and a number of temples and study centers associated with the

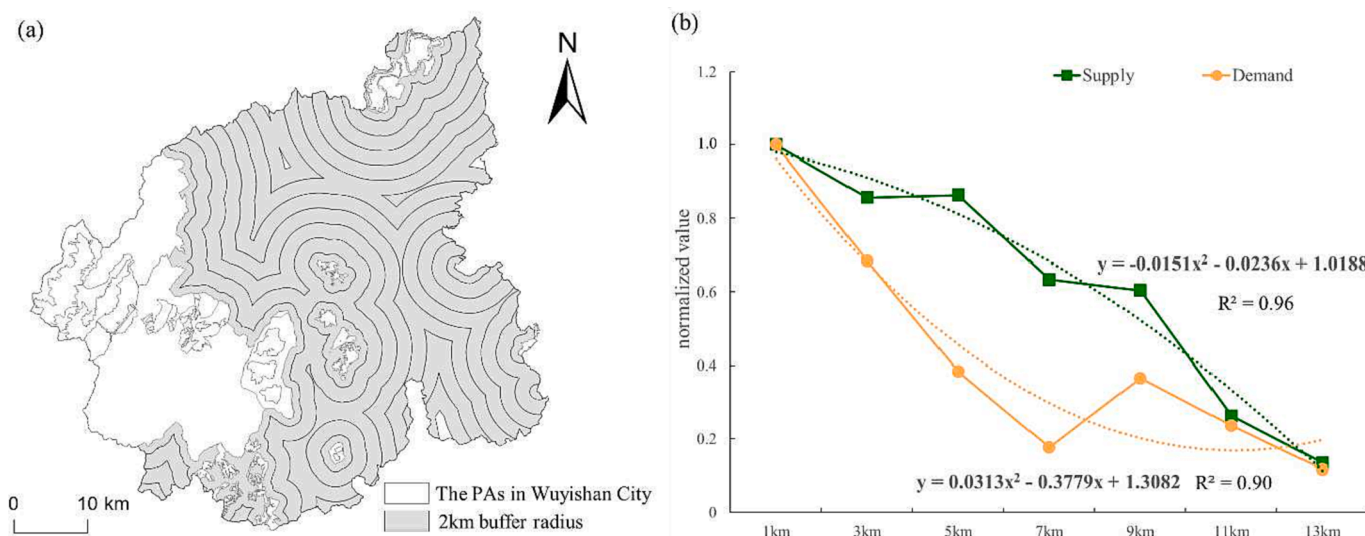


Fig. 8. (a) PA buffer gradient relationship. (b) Dotted lines represent polynomial fit lines.

birth of NeoConfucianism in the 11th century CE, and is more developed, while the northern region is dominated by primary and secondary industries, and has been slow to develop. Heritage sites play an important role in CESs supply (Bachi et al., 2020; Ridding et al., 2018), and areas with better infrastructure will also increase CESs supply (Baumeister et al., 2020). The results were consistent with the conclusions of previous studies (Bachi et al., 2020; Baumeister et al., 2020; Pereira et al., 2021; Ridding et al., 2018).

CES demand is positively correlated with people's knowledge of CESs, and people's perceptions of and access to CESs depend on their knowledge of the region (Casado-Arzuaga et al., 2013; Fei et al., 2016) and the accessibility of destinations (Ala-Hulkko et al., 2016). There are differences in the number and layout of landscape types and the level of economic development in various regions, resulting in a range of people's preference for different types of CESs, and thus to a certain extent there are differences in demand (Fei et al., 2016). In this study, we used social media data to assess the level of public demand for three types of CESs, making full use of the public's various evaluations and ratings of local landscapes, and considered the level of public demand objectively. The results indicated that high demand was concentrated in several hot spots, while demand for CESs is lower than supply in other areas, especially around scenic spots such as WNP. Numerous previous studies have found a positive correlation between CES demand and population density (Fei et al., 2016). Notably, if supply and demand are quantified based on towns in the study area, there may be some discrepancy when using population density (resident population and tourist population) to measure the level of demand for CESs. Specifically, data on the towns' resident population were difficult to obtain because the population yearbook is focused on the county level, with few statistics for individual towns (Qin et al., 2014). It is easy to obtain the tourist population in areas with large numbers of tourists, but it is difficult to calculate the tourist population in areas with small numbers of tourists, and thus there may be inaccuracies in the data.

The mismatch between the supply and demand of ecosystem services may lead to excessive consumption or low utilization of natural resources. The results of supply and demand matching shows that the potential supply is greater than the demand in Wuyishan city, and thus there is large potential for development in the future. Although the supply–demand ratio was used to measure the CESs surplus and deficit in the study area, this indicator may have some limitations due to the complexity of human–environment systems, as well as data and methodological uncertainties. Despite those limitations, this supply–demand ratio index has been widely and successfully applied in many case

studies as the strengths of quantifying and simplicity (Li et al., 2022).

Generally, supply is more studied than demand in previous studies (Kienast et al., 2009; Liu et al., 2021). The spatial quantization methods of demand are still limited. For instance, the questionnaire-based method (Peña et al., 2015) and expert experience method (Burkhard et al., 2012a) fall short in providing spatial quantification. The SolVES model has provided a high level of accuracy for the evaluation of CES supply in previous studies (Zhao et al., 2019). The index can evaluate the value of various CESs in a non-monetized form, and can intuitively display the overall distribution of CES values and the location of high and low values (Zhang et al., 2021). However, traditional questionnaires are still used to obtain social survey data. It is worth discussion to identify better and more objective data acquisition and conversion methods to reduce bias. In addition, appropriate environmental data should be selected according to different regional characteristics to fully reflect the real situation of the study area.

4.2. Strengthening the evaluation and protection of ecosystem services inside and outside PAs

PAs provide high-value CESs in the ecosystem (Tengberg et al., 2012). The total supply and demand values inside PAs are far greater than those outside PAs, and the supply of CESs exceeds demand in all PAs, which can meet people's demand for CESs (Maldonado et al., 2019). The particularity of the landscape can affect people's perceptions of CESs (Iverson Nassauer, 1995). Many PAs are characterized by multiple types and functions. For example, the WCNH, which includes WNP, has the main functions of protecting the ecological environment and maintaining the ecosystem balance (Sheng et al., 2020), but it also plays the roles of developing scenic resources and promoting social and economic development (Chen et al., 2021). Thus, it is a complex system of intertwined influences of both nature and society. This requires the region to combine protection and development to maintain the sustainable development of PAs.

Cultural diversity depends on the diversity of local landscapes, and the loss of cultural identity is associated with the loss of specific PAs (Tengberg et al., 2012). Previous studies have recognized the importance of cultural heritage and cultural diversity in protecting and delivering cultural and ecological values (Tengberg et al., 2012). For example, Sweden and other European countries maintain areas of high-value cultural and natural heritage in a special way (Hasund, 2009). Since the establishment of the first nature reserve in 1956, numerous PAs have been established in China, covering forests, grasslands,

wetlands, deserts, and marine ecosystems, as well as rare and endangered species and germplasm resources, geological relics, natural landscapes, and important water sources. PAs enable long-term conservation of the ecosystem services and cultural values that natural resources hold. The conservation and inheritance innovation of CES in PAs cannot be ignored for sustainable ecosystem management. There are various natural PAs in Wuyishan City, but some of them overlap with each other and there are local differences. Different types of PAs provide different ecosystem services (Potts et al., 2014), and thus it will be necessary to comprehensively evaluate and optimize the integration of the overlapping hot spots within PAs based on their dominant ecosystem services in the future. Moreover, the protection of PAs cannot be limited to management within the PAs. It is also necessary to enhance natural and cultural diversity beyond the PAs to achieve joint development, especially in areas with a relatively large gap between supply and demand of CESs outside the PA.

4.3. CES supply and demand and landscape governance

The supply and demand of CESs are closely related to landscape characteristics (Plieninger et al., 2013), and people's demand for CESs will increase in the future, which will necessitate the development of CES that are diversified and multidimensional (Xiao et al., 2017). Landscape governance requires the building of interconnections between social spaces and the biophysical conditions of places (Görg, 2007) by strengthening landscape services, using the landscape as a social-ecological network connection point, and using it as a bridge connecting society and nature (Westerink et al., 2017). There is evidence that CES can inform landscape governance and incentivize landscape versatility (Chan et al., 2012). Landscape governance integrating with CESs can reveal the material and non-material benefits that stakeholders attach to local landscape, especially the social values that have been overlooked by biophysical and economic monetary assessments (Plieninger et al., 2015). We need to continue to tap into the inherent potential of the landscape so that people can benefit from it to enhance human well-being and achieve CES management goals such as win-win synergy between ecosystem services and sub-services within CESs.

The CES supply capacity is greater than the demand capacity in most areas of Wuyishan City. This shows that regions with strong supply capacity have better economic development potential (Zhao et al., 2019). In the future, environmental factors such as topography can be comprehensively considered in the development process of CESs. Specifically, the development concern should be placed in the middle and low altitude areas and the plain gentle slope, following the principle of moderate development. In comparison, the steep, high altitude, and uninhabited mountains and dense forests are designated as ecological conservation areas with other ecosystem services should be emphasized (e.g. soil and water conservation, provision of ecological products, biodiversity maintenance, etc.) (Zhang, 2021). The government should further increase the use of Zhuzi culture, tea culture, religious culture and other intangible cultural heritage in area with the rich cultural and historical resources. Local, county-level governments may encourage the promotion of various forms of cultural industries and foster emerging cultural formats (smart tourism, research tourism, wellness tourism) (Zhao et al., 2022). For instance, stakeholders could develop folk festival activities that integrate viewing and entertainment according to local conditions, combine education humanities with recreation and ecotourism values, which help tourists enhance the travel experience.

The accessibility will also affect the demand for CESs (Baró et al., 2016; Paracchini et al., 2014). Samat (2010) suggested that road networks can support the tourism industry. Transportation companies regarding the bus route to revise and improve it as well as renovate the buses, because the bus route does not presently cover all tourist destinations (Kantawateera et al., 2015). This will save time for tourists and enhance the tourist experience to facilitate the flow of supply-demand

for CES (Ala-Hulkko et al., 2016).

In the development of conservation policies, priority is usually given to areas with low supply and high demand. We need to pay attention to and protect hot spots to ensure that the ecological environment is not overloaded and eventually destroyed. For example, the results indicated that a few areas (the central area of Wuyishan City) are extremely unbalanced in terms of supply and demand. They have obvious demand for CESs, but there are numerous administrative offices in the city center and relatively little infrastructure, which does not give people spiritual satisfaction. Thus, museums, cultural centers, and other cultural places should be established in the city, and various cultural and arts festivals should be regularly held to enrich people's spiritual life.

When we formulate development policies, we should give priority to the development of areas with high supply and low demand, accelerate the construction of the tourism landscape and infrastructure such as transportation and accommodation, and promote demand in these areas (Meng et al., 2020). For example, the WNR and the vicinity of Xingtian Town were typical high-supply and low-demand areas. However, because of natural environmental factors (control and policy restrictions), WNR was unable to accommodate large-scale tourism activities. While Xingtian Town has many historical and cultural relics, the public is not familiar with many locations. Therefore, the local characteristics should be fully explored and the local culture should be promoted. We should give priority to investing in the development of areas with a serious imbalance between supply and demand (e.g., Xinfeng Street and Chong'an Street), and commit to comprehensive development.

CES are increasingly linked to human well-being, and stakeholder preferences are a prerequisite for the effective assessment of the value of ecosystem services (Wang et al., 2017). People's perceptions of and preferences regarding CESs can shape their attitudes and behaviors, and managers should incorporate stakeholders' perceived importance of CESs into their decision-making. We should pay attention to the supply capacity and accessibility of CESs, achieve a sustainable supply of cultural ecology, and improve public support for sustainable ecosystem management (Shi et al., 2020).

5. Conclusions

In this study, we developed a spatially explicit approach to effectively assessing the supply and demand of three CESs, and the matching of supply and demand, in a county in China and identified various hot spots, which provides strong support for landscape governance and regional planning. The supply capacity of most CESs was greater than the demand capacity in the study area. The total supply and demand value inside the PAs is much larger than that outside them, and thus they can meet people's higher demand. Green space and infrastructure construction associated with CES should be strengthened outside the PAs to achieve joint development, especially in areas with a relatively large gap between the supply and demand of CESs.

CRedit authorship contribution statement

Yuqing Zhao: Conceptualization, Methodology, Software, Writing – original draft. **Weibin You:** Conceptualization, Supervision, Writing – original draft. **Xueer Lin:** Methodology, Software. **Dongjin He:** Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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